

A Brief Introduction to Fluid Dynamics

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The continuum hypothesis

Any volume of fluid, no matter how small, can be associated with the same macroscopic properties that we associate with the fluid in bulk.

Remark

A fluid can be considered to be a continuous aggregate of infinitesimal **fluid particles** located at each point, each with associated velocity, temperature, etc.

Fluid Particles

The continuum hypothesis

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Length scales

If L_1 is the length scale related to the average distance between neighboring molecules and L_3 is a length scale associated with the macroscopic flow, the continuum hypothesis is valid if there exists a length scale L_2 such that $L_1 \ll L_2 \ll L_3$.

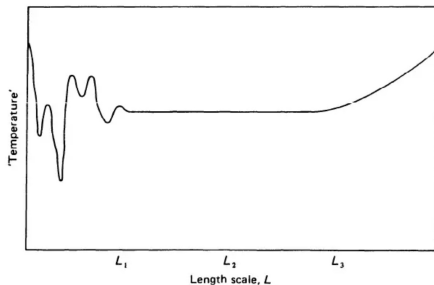


FIG. 5.1 Schematic variation of average energy of molecules with length scale.

Credit: D.J. Tritton, "Physical Fluid Dynamics"

Conservation of Mass

- $\mathbf{u} = \mathbf{u}(\mathbf{r}, t) = (u, v, w)$ is the velocity of the fluid at a given point and time
- ρ is the mass density of the fluid

If V is an arbitrary volume fixed in our coordinate system, the rate of mass loss from V is $\oint_S \rho \mathbf{u} \cdot d\mathbf{A}$. The total mass in V is $\int_V \rho dV$, so if mass is conserved,

$$\begin{aligned}\frac{d}{dt} \int_V \rho dV &= \int_V \frac{\partial \rho}{\partial t} dV = - \oint_S \rho \mathbf{u} \cdot d\mathbf{A} = - \int_V (\nabla \cdot \rho \mathbf{u}) dV \\ \implies \frac{\partial \rho}{\partial t} &= -\nabla \cdot \rho \mathbf{u} \quad (\text{continuity equation})\end{aligned}$$

If ρ is constant, then

$$\nabla \cdot \mathbf{u} = 0$$

The Substantive Derivative

Newton's Second Law

The rate of change of momentum of an object is equal to the net force acting on it.

The rate of change of a quantity T following a fluid particle is

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z}$$

The operator, known as the substantive derivative, can be written as

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla$$

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Rate of change of momentum

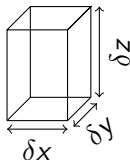
We can also use the substantive derivative on a vector quantity $\mathbf{B}$, where $(\mathbf{u} \cdot \nabla)\mathbf{B}$ is defined through its Cartesian expansion. The case $\mathbf{B} = \mathbf{u}$ give us the rate of change of velocity following a fluid particle,

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial\mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla\mathbf{u}$$

So the rate of change of momentum per unit volume following the fluid is

$$\rho \frac{D\mathbf{u}}{Dt} = \rho \frac{\partial\mathbf{u}}{\partial t} + \rho \mathbf{u} \cdot \nabla\mathbf{u}$$

Pressure force



Consider a rectangular volume element with dimensions δx , δy , δz . Pressure is force per unit area, so

- Force on the left face (pointing right): $p(x, y, z)\delta y\delta z$
- Force on the right face (pointing left): $p(x + \delta x, y, z)\delta y\delta z$
- Net force in the x-direction:

$$p(x, y, z)\delta y\delta z - p(x + \delta x, y, z)\delta y\delta z = -\frac{\partial p}{\partial x}\delta x\delta y\delta z$$

Repeating this for each face, the net force per unit volume is

$$-\frac{\partial p}{\partial x} - \frac{\partial p}{\partial y} - \frac{\partial p}{\partial z} = -\nabla p$$

Stresses and Strains

Definition

A **stress** is a force per unit area across a boundary, for any boundary orientation.

Definition

Strain is a measure of material deformation, expressed as the ratio of the change in length to the original length.

Viscous stresses

Viscous stresses oppose relative movement between neighboring fluid particles, or equivalently, the deformation of the fluid.

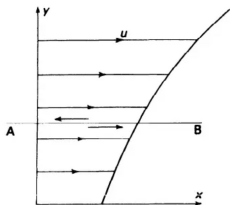


FIG. 1.1 Schematic diagram of viscous stress generated by simple velocity variation, $u(y)$. Short arrows represent forces acting in plane of AB, drawn on side of fluid on which force acts.

Credit: D.J. Tritton, "Physical Fluid Dynamics"

We expect the rate of deformation (rate of strain) to be related to the velocity gradients in the fluid.

Rate of Strain

By considering the separation δl between a pair of fluid particles and expanding $\frac{D(\delta l)^2}{Dt}$, one finds that the sources of strain in the fluid are the symmetrical combinations of the velocity gradients $(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})$. Assuming that the viscous stress tensor and the rate-of-strain tensor are linearly related, the viscous stress tensor is given by

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \delta_{ij} \nabla \cdot \mathbf{u}$$

The net viscous force on a fluid particle is the difference in stresses acting on opposite faces. Thus the force per unit volume is $\sum_j \frac{\partial \tau_{ij}}{\partial x_j}$.

The Navier-Stokes equations

The x -component of the viscous force per unit volume is

$$\frac{\partial}{\partial x} \left[2\mu \frac{\partial u}{\partial x} + \lambda \nabla \cdot \mathbf{u} \right] + \frac{\partial}{\partial y} \left[\mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \right] + \frac{\partial}{\partial z} \left[\mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right]$$

If ρ is constant, then $\nabla \cdot \mathbf{u} = 0$. If μ is also constant, this expression simplifies to $\mu \nabla^2 u$. Collecting together the various contributions, we obtain

The Navier-Stokes equations

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{F}$$

where $\mathbf{F}$ denotes any external body forces, and $\nabla^2 \mathbf{u}$ is defined through its Cartesian expansion.

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Dynamical similarity

Central question

Under what conditions do similar flow patterns occur in two geometrically similar pieces of apparatus?

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Dynamical similarity (cont.)

We first introduce length and velocity scales, L and U , typical of the size of the apparatus and the rate of fluid flow through it. Considering the Navier-Stokes equations in Cartesian coordinates, we use these scales to non-dimensionalize the relevant variables:

$$\begin{aligned}x' &= x/L, & y' &= y/L & u' &= u/U & v' &= v/U \\z' &= z/L, & t' &= tU/L & w' &= w/U & (\Delta p)' &= \Delta p/\rho U^2\end{aligned}$$

Substituting these non-dimensional forms into the Navier-Stokes and continuity equations gives

$$\nabla' \cdot \mathbf{u}' = 0$$

$$\frac{\partial \mathbf{u}'}{\partial t'} + \mathbf{u}' \cdot \nabla' \mathbf{u}' = -\nabla'(\Delta p)' + \frac{\mu}{\rho UL} \nabla'^2 \mathbf{u}'$$

Reynolds number

Definition

The **Reynolds number** of a flow is $Re = \frac{UL}{\nu}$, where ν is kinematic viscosity μ/ρ .

$$\nabla' \cdot \mathbf{u}' = 0$$

$$\frac{\partial \mathbf{u}'}{\partial t'} + \mathbf{u}' \cdot \nabla' \mathbf{u}' = -\nabla'(\Delta p)' + \frac{1}{Re} \nabla'^2 \mathbf{u}'$$

Boundary conditions can be similarly non-dimensionalized, so that if two flows in geometrically similar situations have the same Reynolds number, then the equations for the non-dimensional parameters are the same, and the same flow patterns occur.

Physical interpretation of Re

The Navier-Stokes equations for steady incompressible flow are

$$\rho \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}$$

The second and third terms are the pressure force and viscous force, respectively. The first term is referred to as the inertia force. Making some order-of-magnitude arguments, we find

$$|\mathbf{u} \cdot \nabla \mathbf{u}| \sim U^2/L, \quad |\nu \nabla^2 \mathbf{u}| \sim \nu U/L^2$$

so

$$\frac{|\mathbf{u} \cdot \nabla \mathbf{u}|}{|\nu \nabla^2 \mathbf{u}|} \sim \frac{UL}{\nu} = \text{Re}$$

That is, the Reynolds number is roughly the ratio of the inertia force to the viscous force.

Low Reynolds number

When $Re \ll 1$, the inertia force is negligible compared to the viscous force, and we can use the approximation

$$\nabla p = \mu \nabla^2 \mathbf{u} \quad (\text{Stokes equation})$$

with the possible addition of external body forces.

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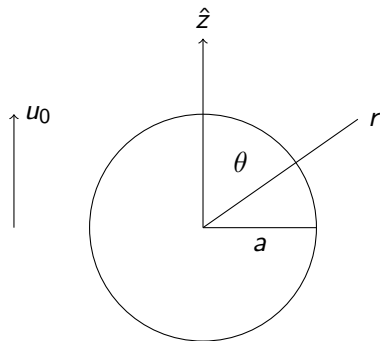
Boundary conditions

Consider an impermeable object with velocity $\mathbf{U}$ and outward normal $\hat{\mathbf{n}}$ placed in a flow with free-stream velocity $\mathbf{u}_0$.

- ① No-slip condition: $\mathbf{u} \times \hat{\mathbf{n}} = \mathbf{U} \times \hat{\mathbf{n}}$
- ② Impermeability: $\mathbf{u} \cdot \hat{\mathbf{n}} = \mathbf{U} \cdot \hat{\mathbf{n}}$
- ③ Infinite boundary: $\mathbf{u} \rightarrow \mathbf{u}_0$ as $\mathbf{r} \rightarrow \infty$

Conditions (1) and (2) give $\mathbf{u} = \mathbf{U}$.

Problem Statement



$$\nabla p = \mu \nabla^2 \mathbf{u}$$

$$\nabla \cdot \mathbf{u} = 0$$

$$u_r = u_\theta = 0 \text{ at } r = a$$

$$u_r \rightarrow u_0 \cos \theta, \quad u_\theta \rightarrow -u_0 \sin \theta$$

$$\text{as } r \rightarrow \infty$$

$$u_\phi = 0, \quad \frac{\partial}{\partial \phi} \rightarrow 0 \quad \text{by symmetry}$$

Solution

In spherical polar coordinates, the continuity and Stokes equations are

$$\frac{\partial u_r}{\partial r} + \frac{2u_r}{r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_\theta \cot \theta}{r} = 0 \quad (\text{continuity equation})$$

$$-\frac{\partial p}{\partial r} + \mu \left(\frac{\partial^2 u_r}{\partial r^2} + \frac{2}{r} \frac{\partial u_r}{\partial r} - \frac{2u_r}{r^2} + \frac{1}{r^2} \frac{\partial^2 u_r}{\partial \theta^2} + \frac{\cot \theta}{r^2} \frac{\partial u_r}{\partial \theta} - \frac{2}{r^2} \frac{\partial u_\theta}{\partial \theta} - \frac{2u_\theta \cot \theta}{r^2} \right) = 0$$

$$-\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left(\frac{\partial^2 u_\theta}{\partial r^2} + \frac{2}{r} \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r^2 \sin^2 \theta} + \frac{1}{r^2} \frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{\cot \theta}{r^2} \frac{\partial u_\theta}{\partial \theta} + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} \right) = 0$$

We assume a solution of the form

$$u_r = R_r(r) \Theta_r(\theta)$$

$$u_\theta = R_\theta(r) \Theta_\theta(\theta)$$

$$\Delta p = R_p(r) \Theta_p(\theta)$$

Solution

The choices $\Theta_r = u_0 \cos \theta$, $\Theta_\theta = -u_0 \sin \theta$, $\Theta_p = \mu u_0 \cos \theta$ satisfy the equations and boundary conditions, leaving us with the system

$$\begin{aligned}\frac{dR_r}{dr} + \frac{2}{r}R_r &= \frac{2}{r}R_\theta \\ \frac{dR_p}{dr} &= \left(\frac{d^2R_r}{dr^2} + \frac{4}{r} \frac{dR_r}{dr} \right) \\ \frac{1}{r}R_p &= \left(\frac{d^2R_\theta}{dr^2} + \frac{2}{r} \frac{dR_\theta}{dr} - \frac{1}{r} \frac{dR_r}{dr} \right)\end{aligned}$$

with boundary conditions

- ① $R_r \rightarrow 1$ and $R_\theta \rightarrow 1$ as $r \rightarrow \infty$
- ② $R_r(a) = R_\theta(a) = 0$

Solution

We assume a solutions of the general form $R_i = \sum_{n=-\infty}^{\infty} a_{n,i} r^n$.

Noting that $a_{n,i} = 0$ for all $n > 0$, we instead write

$$R_r = \sum_{n=0}^{\infty} \frac{a_n}{r^n}, \quad R_\theta = \sum_{n=0}^{\infty} \frac{b_n}{r^n}, \quad R_p = \sum_{n=0}^{\infty} \frac{c_n}{r^n}$$

By equating like powers of $\frac{1}{r}$, we can convert the system of differential equations into an algebraic system.

After some algebra, we obtain

$$u_r = u_0 \cos \theta \left(1 - \frac{3a}{2r} + \frac{1}{2} \frac{a^3}{r^3} \right)$$

$$u_\theta = -u_0 \sin \theta \left(1 - \frac{3a}{4r} - \frac{1}{4} \frac{a^3}{r^3} \right)$$

$$\Delta p = \mu u_0 \cos \theta \left(-\frac{3a}{2r^2} \right)$$

Drag Force

The force per unit area in the flow direction acting at a point on the surface of the sphere is

$$\sigma = -\mu \left(\frac{\partial u_\theta}{\partial r} \right)_{r=a} \sin \theta - (\Delta p)_{r=a} \cos \theta$$

Substituting in the solutions for u_θ and Δp gives

$$\sigma = \frac{3\mu u_0}{2a}$$

Since σ is independent of θ , the total force on the sphere is σ multiplied by the surface area of the sphere:

$$D = 6\pi\mu a u_0$$